MH1201 Linear Algebra II

Midterm Exam (Solutions)

AY 2023/2024, Semester 2

Quantitative Research Society @NTU

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Overview

- Linear transformations between polynomial spaces: linearity, matrix representations w.r.t. standard bases, kernel and range, rank–nullity, injective/surjective tests.
- Subspace tests in matrix spaces: symmetric matrices, affine translates of subspaces, and structural criteria for (non-)subspaces.
- Basis manipulations: verifying a list is a basis via linear independence/spanning and change-of-basis matrices.
- Solution structure: each question includes multiple genuinely distinct methods; final answers are boxed and each question ends with a mark scheme.

Contents

Question 1 (15 marks)

Let $T : P_2(\mathbb{R}) \to P_4(\mathbb{R})$ be the map defined by $T(f(x)) = f(x^2)$.

(a) Show that T is a linear map.

(b) Let $B = \{1, x, x^2\}$ be the standard basis for $P_2(\mathbb{R})$ and $A = \{1, x, x^2, x^3, x^4\}$ be the standard basis for $P_4(\mathbb{R})$. Write down the matrix representation of T with respect to B and A.

(c) Determine a basis for either null(T) OR range(T).

(d) Determine the dimensions of null(T) AND range(T).

(e) Determine whether the following properties of T is true. You are to explain your answer.

(i) Is T injective?

(ii) Is T surjective?

Solution

Method 1: Basis images + matrix rank (standard / explanatory)

(a) **Linearity.** Let $f, g \in P_2(\mathbb{R})$ and $\lambda \in \mathbb{R}$. Then for all $x \in \mathbb{R}$,

$$T(f + \lambda g)(x) = (f + \lambda g)(x^2) = f(x^2) + \lambda g(x^2) = T(f)(x) + \lambda T(g)(x),$$

hence $T(f + \lambda g) = T(f) + \lambda T(g)$. Therefore T is linear.

$$\boxed{T \text{ is linear.}}$$

(b) **Matrix representation.** Compute images of the basis $B = (1, x, x^2)$:

$$T(1) = 1, \qquad T(x) = x^2, \qquad T(x^2) = x^4.$$

With respect to $A = (1, x, x^2, x^3, x^4)$, the coordinate vectors are

$$[T(1)]_A = \begin{bmatrix}1\\0\\0\\0\\0\end{bmatrix}, \quad [T(x)]_A = \begin{bmatrix}0\\0\\1\\0\\0\end{bmatrix}, \quad [T(x^2)]_A = \begin{bmatrix}0\\0\\0\\0\\1\end{bmatrix}.$$

Thus

$$[T]_B^A = \begin{bmatrix}1 & 0 & 0\\0 & 0 & 0\\0 & 1 & 0\\0 & 0 & 0\\0 & 0 & 1\end{bmatrix}.$$

$$\boxed{[T]_B^A = \begin{bmatrix}1 & 0 & 0\\0 & 0 & 0\\0 & 1 & 0\\0 & 0 & 0\\0 & 0 & 1\end{bmatrix}}$$

(c) **A basis for** range(T)**.** Let $f(x) = a + bx + cx^2 \in P_2(\mathbb{R})$. Then

$$T(f)(x) = f(x^2) = a + bx^2 + cx^4.$$

Hence

$$\text{range}(T) = \{a + bx^2 + cx^4 : \ a, b, c \in \mathbb{R}\} = \text{span}\{1, x^2, x^4\}.$$

To show $\{1, x^2, x^4\}$ is linearly independent, suppose

$$\alpha \cdot 1 + \beta x^2 + \gamma x^4 \equiv 0 \quad \text{in } P_4(\mathbb{R}).$$

Comparing coefficients gives $\alpha = \beta = \gamma = 0$. Thus $\{1, x^2, x^4\}$ is a basis of range(T).

$$\boxed{\text{A basis for range}(T) \text{ is } \{1, x^2, x^4\}.}$$

(d) **Dimensions of** null(T) **and** range(T)**.** From (c), $\dim \text{range}(T) = 3$.

Also, the matrix in (b) has 3 pivot columns (equivalently rank 3), so $\text{rank}(T) = 3$. Since $\dim P_2(\mathbb{R}) = 3$, rank–nullity gives

$$\dim \text{null}(T) = 3 - \text{rank}(T) = 3 - 3 = 0.$$

$$\boxed{\dim \text{null}(T) = 0, \qquad \dim \text{range}(T) = 3.}$$

(e) **Injective / surjective.**

(i) T is injective $\iff$ $\text{null}(T) = \{0\}$. Since $\dim \text{null}(T) = 0$, T is injective.

(ii) T is surjective $\iff$ $\text{range}(T) = P_4(\mathbb{R})$. But $\dim \text{range}(T) = 3 < \dim P_4(\mathbb{R}) = 5$, so T is not surjective.

$$\boxed{T \text{ is injective but not surjective.}}$$

Method 2: Coefficient comparison

Write $f(x) = a + bx + cx^2$ and $g(x) = a' + b'x + c'x^2$.

(a) **Linearity.** For all x,

$$T(f + \lambda g)(x) = (f + \lambda g)(x^2) = f(x^2) + \lambda g(x^2) = T(f)(x) + \lambda T(g)(x),$$

so T is linear.

$$\boxed{T \text{ is linear.}}$$

(b) **Matrix representation.** Using $T(1) = 1$, $T(x) = x^2$, $T(x^2) = x^4$ and reading coordinates in A gives

$$\boxed{[T]_B^A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}.$$

(c) **A basis for** null(T) **OR** range(T)**.** If $T(f) = 0$, then

$$0 \equiv T(f)(x) = a + bx^2 + cx^4 \quad \Rightarrow \quad a = b = c = 0$$

by coefficient comparison in $P_4(\mathbb{R})$. Hence $\text{null}(T) = \{0\}$. Also,

$$\text{range}(T) = \{a + bx^2 + cx^4 : \ a, b, c \in \mathbb{R}\} = \text{span}\{1, x^2, x^4\}.$$

$$\boxed{\text{null}(T) = \{0\}, \text{ and } \{1, x^2, x^4\} \text{ is a basis of } \text{range}(T).}$$

(d) **Dimensions.** $\dim \text{null}(T) = 0$ and $\dim \text{range}(T) = 3$ (three free parameters a, b, c).

$$\boxed{\dim \text{null}(T) = 0, \qquad \dim \text{range}(T) = 3.}$$

(e) **Injective / surjective.**

(i) $\text{null}(T) = \{0\} \Rightarrow T$ injective.

(ii) $\dim \text{range}(T) = 3 < 5 \Rightarrow T$ not surjective.

$$\boxed{T \text{ is injective but not surjective.}}$$

Method 3: Even-subspace characterisation via composition

Theorem (Roots of a nonzero polynomial are finite). If $p \in \mathbb{R}[x]$ is not the zero polynomial, then p has only finitely many real roots.

Let $g(x) = x^2$, so $T(f) = f \circ g$.

(a) **Linearity.** For all x,

$$T(f + \lambda h)(x) = (f + \lambda h)(x^2) = f(x^2) + \lambda h(x^2) = T(f)(x) + \lambda T(h)(x),$$

so T is linear.

$$\boxed{T \text{ is linear.}}$$

(b) **Matrix representation.** $T(1) = 1$, $T(x) = x^2$, $T(x^2) = x^4$; hence

$$\boxed{[T]_B^A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}.$$

(c) **Range and kernel.** For any f, $T(f)$ is even since $T(f)(-x) = f((-x)^2) = f(x^2) = T(f)(x)$. Hence range$(T)$ is contained in the even subspace

$$W := \{p \in P_4(\mathbb{R}) : p(-x) = p(x)\} = \operatorname{span}\{1, x^2, x^4\}.$$

Conversely, each $a + bx^2 + cx^4 \in W$ equals $T(a + bx + cx^2)$, so range$(T) = W$.

If $T(f) = 0$, then $f(x^2) = 0$ for all x, so $f(t) = 0$ for all $t \geq 0$ (take $t = x^2$). Since $[0, \infty)$ is infinite, the theorem implies f must be the zero polynomial. Hence null$(T) = \{0\}$.

$$\boxed{\operatorname{range}(T) = \operatorname{span}\{1, x^2, x^4\}, \qquad \operatorname{null}(T) = \{0\}.}$$

(d) **Dimensions.** $\dim \operatorname{range}(T) = 3$ and $\dim \operatorname{null}(T) = 0$.

$$\boxed{\dim \operatorname{null}(T) = 0, \qquad \dim \operatorname{range}(T) = 3.}$$

(e) **Injective / surjective.**

(i) $\operatorname{null}(T) = \{0\} \Rightarrow T$ injective.

(ii) $\operatorname{range}(T) = W \subsetneq P_4(\mathbb{R})$ (e.g. $x \notin W$) $\Rightarrow T$ not surjective.

$$\boxed{T \text{ is injective but not surjective.}}$$

Method 4: Explicit isomorphism onto the even subspace

Define the even subspace

$$W := \operatorname{span}\{1, x^2, x^4\} \subset P_4(\mathbb{R}).$$

Define a linear map

$$S : W \to P_2(\mathbb{R}), \qquad S(a + bx^2 + cx^4) = a + bx + cx^2.$$

(a) **Linearity.** For $f, g \in P_2$ and $\lambda \in \mathbb{R}$,

$$T(f + \lambda g)(x) = (f + \lambda g)(x^2) = f(x^2) + \lambda g(x^2) = T(f)(x) + \lambda T(g)(x),$$

so T is linear.

$$\boxed{T \text{ is linear.}}$$

(b) **Matrix representation.** $T(1) = 1$, $T(x) = x^2$, $T(x^2) = x^4$; thus

$$\boxed{[T]_B^A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}.$$

(c) **Range/kernel via two-sided inverse on W.** For $f(x) = a + bx + cx^2$, we have $T(f) = a + bx^2 + cx^4 \in W$, so $T(P_2) \subseteq W$.

Moreover, for all $f \in P_2$,

$$(S \circ T)(f) = S\big(a + bx^2 + cx^4\big) = a + bx + cx^2 = f,$$

so $S \circ T = \mathrm{Id}_{P_2}$, which implies T is injective and $T(P_2)$ has dimension 3.

Also, for any $w = a + bx^2 + cx^4 \in W$,

$$(T \circ S)(w) = T(a + bx + cx^2) = a + bx^2 + cx^4 = w,$$

so $T \circ S = \mathrm{Id}_W$, hence $T(P_2) = W$ and T restricts to an isomorphism $P_2 \cong W$. Therefore a basis of range(T) is $\{1, x^2, x^4\}$ and null$(T) = \{0\}$.

$$\boxed{\operatorname{range}(T) = W = \operatorname{span}\{1, x^2, x^4\}, \qquad \operatorname{null}(T) = \{0\}.}$$

(d) **Dimensions.** Since $T : P_2 \to W$ is an isomorphism, $\dim \operatorname{range}(T) = \dim W = 3$ and $\dim \operatorname{null}(T) = 0$.

$$\boxed{\dim \operatorname{null}(T) = 0, \qquad \dim \operatorname{range}(T) = 3.}$$

(e) **Injective / surjective.**

(i) Injective since $S \circ T = \mathrm{Id}_{P_2}$.

(ii) Not surjective onto P_4 since $\operatorname{range}(T) = W \subsetneq P_4(\mathbb{R})$.

$$\boxed{T \text{ is injective but not surjective.}}$$

Method 5: Rank by a nonzero minor + rank–nullity

Theorem (Rank from a nonzero minor). A matrix has rank at least r if it contains an $r \times r$ minor with nonzero determinant.

Theorem (Rank–Nullity). For linear $T : V \to W$ with $\dim V < \infty$,

$$\dim V = \dim \operatorname{null}(T) + \dim \operatorname{range}(T).$$

(a) **Linearity.** As in previous methods, $T(f + \lambda g)(x) = (f + \lambda g)(x^2) = f(x^2) + \lambda g(x^2) = T(f)(x) + \lambda T(g)(x)$, hence T linear.

$$\boxed{T \text{ is linear.}}$$

(b) **Matrix representation.** $T(1) = 1$, $T(x) = x^2$, $T(x^2) = x^4$, so

$$\boxed{[T]_B^A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.}$$

(c) **Basis for** $\operatorname{range}(T)$**.** From $T(a + bx + cx^2) = a + bx^2 + cx^4$, we have $\operatorname{range}(T) = \operatorname{span}\{1, x^2, x^4\}$, and coefficient comparison shows independence, hence it is a basis.

$$\boxed{\text{A basis for } \operatorname{range}(T) \text{ is } \{1, x^2, x^4\}.}$$

(d) **Dimensions.** In $[T]_B^A$, consider the 3×3 minor using rows $1, 3, 5$ and all 3 columns:

$$\det \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = 1 \neq 0.$$

Thus $\operatorname{rank}(T) = 3$. Since $\dim P_2 = 3$, rank–nullity gives $\dim \operatorname{null}(T) = 0$ and $\dim \operatorname{range}(T) = 3$.

$$\boxed{\dim \operatorname{null}(T) = 0, \qquad \dim \operatorname{range}(T) = 3.}$$

(e) **Injective / surjective.**

(i) $\dim \operatorname{null}(T) = 0 \Rightarrow T$ injective.

(ii) $\dim \operatorname{range}(T) = 3 < 5 \Rightarrow T$ not surjective.

$$\boxed{T \text{ is injective but not surjective.}}$$

Mark Scheme (15 marks)

- **(a) (2 marks)**
  - (1) Computes $T(f+\lambda g)(x)$ and expands using evaluation at x^2.
  - (1) Concludes $T(f+\lambda g) = T(f) + \lambda T(g)$ and states linearity.
- **(b) (4 marks)**
  - (1) Computes $T(1) = 1$ and writes its coordinates in A.
  - (1) Computes $T(x) = x^2$ and writes its coordinates in A.
  - (1) Computes $T(x^2) = x^4$ and writes its coordinates in A.
  - (1) Forms $[T]_B^A$ with correct column order matching B.
- **(c) (3 marks)**
  - (1) Correctly characterises range(T) (or null(T)) from $T(a+bx+cx^2) = a+bx^2+cx^4$.
  - (1) Produces a correct spanning set for the chosen space (e.g. $\{1, x^2, x^4\}$ for range(T)).
  - (1) Justifies that it is a basis (coefficient/independence or dimension argument).
- **(d) (2 marks)**
  - (1) Determines $\dim \operatorname{range}(T) = 3$ (basis or rank).
  - (1) Determines $\dim \operatorname{null}(T) = 0$ (kernel computation or rank–nullity) and states both.
- **(e) (4 marks)**
  - (2) Injective: uses $\operatorname{null}(T) = \{0\}$ criterion (or $\dim(T) = 0$) and concludes injective.
  - (2) Surjective: uses dimension/range obstruction ($\dim \operatorname{range}(T) < 5$ or $x \notin \operatorname{range}(T)$) and concludes not surjective.

Question 2 (10 marks)

Recall that $M_{24\times 24}(\mathbb{R})$ is the vector space comprising all (24×24)-matrices. Also, recall that $S_{24\times 24}(\mathbb{R})$ is the vector space comprising all (24×24)-symmetric matrices.

(a) Let $E_{1,1}$ be the matrix with one at the first row and the first column, and zero elsewhere. In other words, $E_{1,1}$ has exactly one 1. Consider the set $U = \{A + E_{1,1} \in M_{24\times 24}(\mathbb{R}) : A \in S_{24\times 24}(\mathbb{R})\}$. Is U a subspace of $M_{24\times 24}(\mathbb{R})$?

(b) Let $E_{20,24}$ be the matrix with one at the 20-th row and the 24-th column, and zero elsewhere. In other words, $E_{20,24}$ has exactly one 1. Consider the set $V = \{A + E_{20,24} \in M_{24\times 24}(\mathbb{R}) : A \in S_{24\times 24}(\mathbb{R})\}$. Is V a subspace of $M_{24\times 24}(\mathbb{R})$?

Solution

Method 1: Direct identification / zero test (standard / explanatory)

(a) U **is a subspace.** $E_{1,1}$ is symmetric, hence $E_{1,1} \in S_{24\times 24}(\mathbb{R})$. We show $U = S_{24\times 24}(\mathbb{R})$.

If $B \in U$, then $B = A + E_{1,1}$ for some symmetric A. Since $E_{1,1}$ is symmetric, B is a sum of symmetric matrices, hence symmetric; so $B \in S_{24\times 24}(\mathbb{R})$. Conversely, if $B \in S_{24\times 24}(\mathbb{R})$, then $B = (B - E_{1,1}) + E_{1,1}$ and $B - E_{1,1}$ is symmetric, so $B \in U$. Thus $U = S_{24\times 24}(\mathbb{R})$, which is a subspace of $M_{24\times 24}(\mathbb{R})$.

$$\boxed{U \text{ is a subspace of } M_{24\times 24}(\mathbb{R}).}$$

(b) V **is not a subspace.** If V were a subspace, then $0 \in V$. That would mean $A + E_{20,24} = 0$ for some symmetric A, i.e. $A = -E_{20,24}$. But $E_{20,24}^{\mathsf{T}} = E_{24,20} \neq E_{20,24}$, so $-E_{20,24}$ is not symmetric. Hence $0 \notin V$ and V is not a subspace.

$$\boxed{V \text{ is not a subspace of } M_{24\times 24}(\mathbb{R}).}$$

Method 2: Affine-translate criterion

Theorem (Translate of a subspace). Let $W \leq X$ be a subspace and $w_0 \in X$. Then $w_0 + W := \{w_0 + w : \ w \in W\}$ is a subspace of X if and only if $w_0 \in W$. (In that case, $w_0 + W = W$.)

(a) Take $X = M_{24\times 24}(\mathbb{R})$, $W = S_{24\times 24}(\mathbb{R})$, $w_0 = E_{1,1}$. Since $E_{1,1}$ is symmetric, $w_0 \in W$, hence $U = w_0 + W$ is a subspace (indeed $U = W$).

$$\boxed{U \text{ is a subspace of } M_{24\times 24}(\mathbb{R}).}$$

(b) Here $w_0 = E_{20,24} \notin W$ (not symmetric). Hence $V = w_0 + W$ is not a subspace.

$$\boxed{V \text{ is not a subspace of } M_{24\times 24}(\mathbb{R}).}$$

Method 3: Symmetric–skew decomposition

Theorem (Symmetric–skew decomposition). Every $X \in M_{n\times n}(\mathbb{R})$ decomposes uniquely as

$$X = \underbrace{\frac{X + X^{\mathsf{T}}}{2}}_{\text{symmetric}} + \underbrace{\frac{X - X^{\mathsf{T}}}{2}}_{\text{skew-symmetric}}.$$

In particular, X is symmetric iff $\frac{X-X^{\mathsf{T}}}{2} = 0$.

(a) Since $E_{1,1}^{\mathsf{T}} = E_{1,1}$, its skew-symmetric part is 0. If A is symmetric, then

$$\frac{(A + E_{1,1}) - (A + E_{1,1})^{\mathsf{T}}}{2} = \frac{A - A^{\mathsf{T}}}{2} + \frac{E_{1,1} - E_{1,1}^{\mathsf{T}}}{2} = 0.$$

Hence every element of U is symmetric. Conversely, any symmetric B can be written as $(B - E_{1,1}) + E_{1,1}$ with $B - E_{1,1}$ symmetric, so U is exactly the symmetric subspace. Therefore U is a subspace.

$$\boxed{U \text{ is a subspace of } M_{24\times 24}(\mathbb{R}).}$$

(b) For $E_{20,24}$,

$$\frac{E_{20,24} - E_{20,24}^{\mathsf{T}}}{2} = \frac{E_{20,24} - E_{24,20}}{2} \neq 0.$$

If A is symmetric then the skew part of $A + E_{20,24}$ equals the skew part of $E_{20,24}$ (since the skew part of A is 0), hence is nonzero. In particular, 0 (whose skew part is 0) cannot lie in V, so V is not a subspace.

$$\boxed{V \text{ is not a subspace of } M_{24\times 24}(\mathbb{R}).}$$

Method 4: Kernel / preimage under the skew-projection

Define the linear map (skew-projection)

$$\pi_{\text{skew}} : M_{24\times 24}(\mathbb{R}) \to M_{24\times 24}(\mathbb{R}), \qquad \pi_{\text{skew}}(X) = \frac{X - X^{\mathsf{T}}}{2}.$$

Theorem. The kernel of a linear map is a subspace.

Note that

$$\ker(\pi_{\text{skew}}) = \{X : X = X^{\mathsf{T}}\} = S_{24\times 24}(\mathbb{R}).$$

(a) Since $E_{1,1} \in \ker(\pi_{\text{skew}})$, we have

$$U = E_{1,1} + \ker(\pi_{\text{skew}}) = \ker(\pi_{\text{skew}})$$

(because adding an element of the kernel does not change the coset). Hence U is a subspace.

$$\boxed{U \text{ is a subspace of } M_{24\times 24}(\mathbb{R}).}$$

(b) Since $E_{20,24} \notin \ker(\pi_{\text{skew}})$, the coset

$$V = E_{20,24} + \ker(\pi_{\text{skew}})$$

does not contain 0, hence cannot be a subspace. (Equivalently, if it were a subspace it must equal the kernel itself, forcing $E_{20,24} \in \ker(\pi_{\text{skew}})$, contradiction.)

$$\boxed{V \text{ is not a subspace of } M_{24\times 24}(\mathbb{R}).}$$

Method 5: Quotient-space / coset argument

Let $M := M_{24\times 24}(\mathbb{R})$ and $S := S_{24\times 24}(\mathbb{R})$.

Theorem (Cosets in a quotient). If $S \le M$ is a subspace, then the quotient M/S is a vector space. A coset $m + S$ is a subspace of M if and only if $m + S = S$ (equivalently $m \in S$).

(a) $U = E_{1,1} + S$. Since $E_{1,1}$ is symmetric, $E_{1,1} \in S$, hence $E_{1,1} + S = S$. Therefore U is a subspace.

$$\boxed{U \text{ is a subspace of } M_{24\times 24}(\mathbb{R}).}$$

(b) $V = E_{20,24} + S$. Since $E_{20,24}$ is not symmetric, $E_{20,24} \notin S$, hence $E_{20,24} + S \neq S$. Therefore V is not a subspace.

$$\boxed{V \text{ is not a subspace of } M_{24\times 24}(\mathbb{R}).}$$

Mark Scheme (10 marks)

- **(a) (5 marks)**
  - (1) Notes $E_{1,1}$ is symmetric and $S_{24\times 24}(\mathbb{R})$ is a subspace of $M_{24\times 24}(\mathbb{R})$.
  - (2) Shows $U \subseteq S_{24\times 24}(\mathbb{R})$ (sum of symmetric matrices is symmetric).
  - (1) Shows $S_{24\times 24}(\mathbb{R}) \subseteq U$ by writing $B = (B - E_{1,1}) + E_{1,1}$ with $B - E_{1,1}$ symmetric.
  - (1) Concludes U is a subspace (equivalently $U = S_{24\times 24}(\mathbb{R})$).
- **(b) (5 marks)**
  - (1) States a necessary subspace condition (e.g. $0 \in V$) or applies translate/coset criterion.
  - (2) Sets up $0 \in V \Rightarrow A = -E_{20,24}$ with A required to be symmetric.
  - (1) Observes $E_{20,24}^{\mathsf{T}} = E_{24,20} \neq E_{20,24}$, hence $-E_{20,24}$ is not symmetric.
  - (1) Concludes $0 \notin V$ and hence V is not a subspace.

Question 3 (5 marks)

Let $[v_1, v_2, v_3]$ be a basis for the vector space V. Show that $[v_1 + \lambda v_3, v_2 + \mu v_3, v_3]$ is a basis for V for any real values λ and μ.

Solution

Method 1: Linear independence in the given basis (standard / explanatory)

Since $[v_1, v_2, v_3]$ is a basis, $\{v_1, v_2, v_3\}$ is linearly independent.

Let

$$\alpha(v_1 + \lambda v_3) + \beta(v_2 + \mu v_3) + \gamma v_3 = 0.$$

Expanding,

$$\alpha v_1 + \beta v_2 + (\alpha\lambda + \beta\mu + \gamma)v_3 = 0.$$

By linear independence of $\{v_1, v_2, v_3\}$,

$$\alpha = 0, \qquad \beta = 0, \qquad \alpha\lambda + \beta\mu + \gamma = \gamma = 0.$$

Thus $\{v_1 + \lambda v_3,\ v_2 + \mu v_3,\ v_3\}$ is linearly independent. Since it has 3 vectors in a 3-dimensional space, it is a basis.

$$\boxed{[v_1 + \lambda v_3, v_2 + \mu v_3, v_3] \text{ is a basis for } V \text{ for all } \lambda, \mu \in \mathbb{R}.}$$

Method 2: Spanning by recovering the original basis

Let

$$w_1 = v_1 + \lambda v_3, \qquad w_2 = v_2 + \mu v_3, \qquad w_3 = v_3.$$

Then

$$v_3 = w_3, \qquad v_1 = w_1 - \lambda w_3, \qquad v_2 = w_2 - \mu w_3.$$

Hence $\{v_1, v_2, v_3\} \subseteq \operatorname{span}\{w_1, w_2, w_3\}$, so

$$V = \operatorname{span}\{v_1, v_2, v_3\} \subseteq \operatorname{span}\{w_1, w_2, w_3\} \subseteq V,$$

and therefore $\operatorname{span}\{w_1, w_2, w_3\} = V$. Since there are 3 spanning vectors and $\dim V = 3$, $\{w_1, w_2, w_3\}$ is a basis.

$$\boxed{[v_1 + \lambda v_3, v_2 + \mu v_3, v_3] \text{ is a basis for } V.}$$

Method 3: Change-of-basis matrix and determinant (matrix method)

Relative to the basis (v_1, v_2, v_3), the coordinate columns of

$$w_1 = v_1 + \lambda v_3, \qquad w_2 = v_2 + \mu v_3, \qquad w_3 = v_3$$

are

$$[w_1]_{(v_1,v_2,v_3)} = \begin{bmatrix} 1 \\ 0 \\ \lambda \end{bmatrix}, \quad [w_2]_{(v_1,v_2,v_3)} = \begin{bmatrix} 0 \\ 1 \\ \mu \end{bmatrix}, \quad [w_3]_{(v_1,v_2,v_3)} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Thus the change-of-basis matrix is

$$C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \lambda & \mu & 1 \end{bmatrix},$$

which is lower triangular with $\det(C) = 1 \neq 0$. Hence C is invertible, so (w_1, w_2, w_3) is a basis of V.

$$\boxed{[v_1 + \lambda v_3, v_2 + \mu v_3, v_3] \text{ is a basis for } V.}$$

Method 4: Elementary basis transformation theorem

Theorem (Elementary basis transformations). If $(u_1, \ldots, u_n)$ is a basis, then replacing some u_i by $u_i + \alpha u_j$ $(i \neq j)$ yields another basis.

Let (v_1, v_2, v_3) be a basis. Replace v_1 by $v_1 + \lambda v_3$; since this is of the form $u_1 + \alpha u_3$, the theorem implies $(v_1 + \lambda v_3,\ v_2,\ v_3)$ is a basis. Next replace v_2 by $v_2 + \mu v_3$ in that basis; again the theorem applies, giving $(v_1 + \lambda v_3,\ v_2 + \mu v_3,\ v_3)$ as a basis.

$$\boxed{[v_1 + \lambda v_3, v_2 + \mu v_3, v_3] \text{ is a basis for } V.}$$

Method 5: Automorphism on coordinate space)

Let $\Phi : \mathbb{R}^3 \to V$ be the coordinate isomorphism associated with the basis (v_1, v_2, v_3):

$$\Phi(a, b, c) = av_1 + bv_2 + cv_3.$$

Theorem. If Φ is an isomorphism and $C \in GL_3(\mathbb{R})$, then $\{\Phi(Ce_1), \Phi(Ce_2), \Phi(Ce_3)\}$ is a basis of V.

Take

$$C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \lambda & \mu & 1 \end{bmatrix},$$

which satisfies $\det(C) = 1 \neq 0$, hence $C \in GL_3(\mathbb{R})$. Compute:

$$Ce_1 = \begin{bmatrix} 1 \\ 0 \\ \lambda \end{bmatrix}, \quad Ce_2 = \begin{bmatrix} 0 \\ 1 \\ \mu \end{bmatrix}, \quad Ce_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Therefore,

$$\Phi(Ce_1) = v_1 + \lambda v_3, \qquad \Phi(Ce_2) = v_2 + \mu v_3, \qquad \Phi(Ce_3) = v_3,$$

and the theorem implies these form a basis of V.

$$\boxed{[v_1 + \lambda v_3, v_2 + \mu v_3, v_3] \text{ is a basis for } V.}$$

Mark Scheme (5 marks)

- **(5 marks)**
  - (1) Uses that $[v_1, v_2, v_3]$ is a basis $\Rightarrow \{v_1, v_2, v_3\}$ is linearly independent and $\dim V = 3$.
  - (2) Sets up $\alpha(v_1 + \lambda v_3) + \beta(v_2 + \mu v_3) + \gamma v_3 = 0$ and expands in $\{v_1, v_2, v_3\}$.
  - (1) Applies linear independence (or invertibility/determinant criterion) to conclude $\alpha = \beta = \gamma = 0$ (or establishes spanning with 3 vectors).
  - (1) Concludes the new list is a basis of V for all $\lambda, \mu \in \mathbb{R}$.